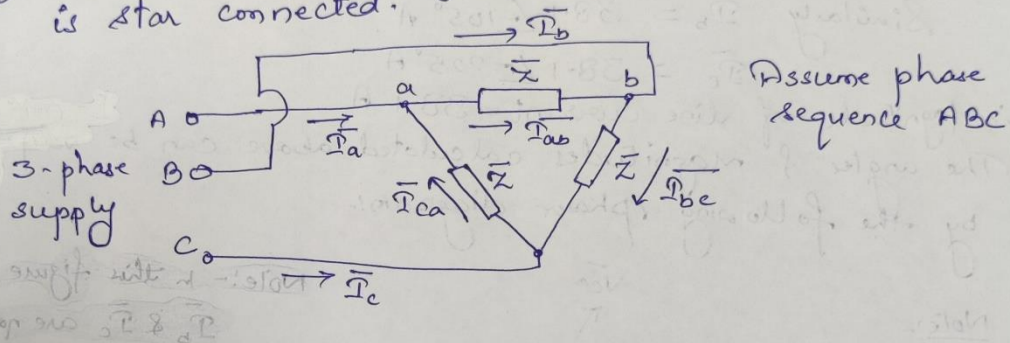


Star and delta connected circuits

①

Q.1

- A) A three-phase 220V source feeds a delta connected three-phase load where per phase load impedance is $10 \angle -45^\circ \Omega$. Find the RMS phase and line currents and draw a labeled phasor diagram. [Refer figure shown below].
- B) Solve the problem given in (A) above, if the load is star connected.



Solution-1:

- A) Take $\bar{V}_{ab}$ as reference phasor

$$\Rightarrow \bar{V}_{ab} = 220 \angle 0^\circ \text{ V}, \quad \bar{V}_{bc} = 220 \angle -120^\circ \text{ V} \text{ \& } \bar{V}_{ca} = 220 \angle -240^\circ \text{ V}$$

$$\bar{I}_{ab} = \frac{\bar{V}_{ab}}{\bar{Z}_{ab}} = \frac{220 \angle 0^\circ}{10 \angle -45^\circ} = 22 \angle 45^\circ \text{ A}$$

$$\text{Similarly } \bar{I}_{bc} = \frac{\bar{V}_{bc}}{\bar{Z}_{bc}} = \frac{220 \angle -120^\circ}{10 \angle -45^\circ} = 22 \angle -75^\circ \text{ A}$$

$$\text{ \& } \bar{I}_{ca} = \frac{\bar{V}_{ca}}{\bar{Z}_{ca}} = \frac{220 \angle -240^\circ}{10 \angle -45^\circ} = 22 \angle -195^\circ \text{ A}$$

Note: $\bar{I}_{bc}$ \& $\bar{I}_{ca}$ can also be directly computed from $\bar{I}_{ab}$ by considering phase shift of 120° , i.e. if $\bar{I}_{ab} = 22 \angle 45^\circ \text{ A}$

$$\Rightarrow \bar{I}_{bc} = 22 \angle 45^\circ - 120^\circ = 22 \angle -75^\circ \text{ A}$$

$$\text{ \& } \bar{I}_{ca} = 22 \angle 45^\circ - 240^\circ = 22 \angle -195^\circ \text{ A}$$

$\therefore$ Magnitude of phase current (RMS) = 22 A.

Calculation of line currents

Apply KCL at Node 'a'

$$\bar{I}_a + \bar{I}_{ca} - \bar{I}_{ab} = 0$$

$$\Rightarrow \bar{I}_a = \bar{I}_{ab} - \bar{I}_{ca} = 22 \angle 45^\circ - 22 \angle -195^\circ$$

$$= 38.1 \angle 15^\circ \text{ A}$$

Similarly $\bar{I}_b = 38.1 \angle -105^\circ \text{ A}$

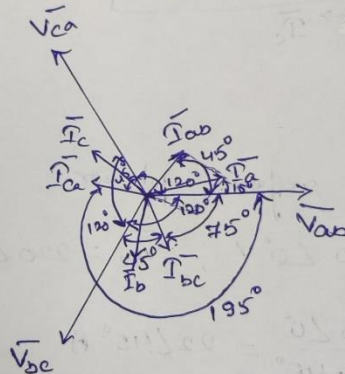
$$\bar{I}_c = 38.1 \angle -225^\circ \text{ A}$$

$\therefore$ Magnitude of line current = 38.1 A [3 times phase current magnitude]

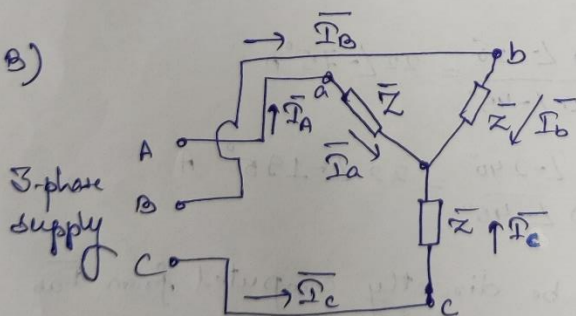
The angles & magnitudes calculated above can be verified by the following phasor diagram:-

Note:-

Angles & lengths are not to scale in this figure



B)



In star connected load line current is the same as phase current. Therefore magnitude of phase & line current is the same.

In star connection, phase voltage is impressed across the load in each phase.

Therefore, first calculate the magnitude of phase voltage, i.e. $|V_p| = |V_a| = |V_b| = |V_c| = \frac{|V_{l-l}|}{\sqrt{3}} = \frac{220 \text{ V}}{\sqrt{3}} = 127 \text{ V}$

Take the 'a' phase voltage, i.e. $\bar{V}_{an}$ (or simply $\bar{V}_a$) as reference, i.e. $\bar{V}_a = 127 \angle 0^\circ \text{ V}$, $\bar{V}_b = 127 \angle -120^\circ \text{ V}$, & $\bar{V}_c = 127 \angle -240^\circ \text{ V}$. ③

$$\text{Current } \bar{I}_a = \frac{\bar{V}_a}{\bar{Z}_a} = \frac{127 \angle 0^\circ}{10 \angle -45^\circ} = 12.7 \angle 45^\circ \text{ A}$$

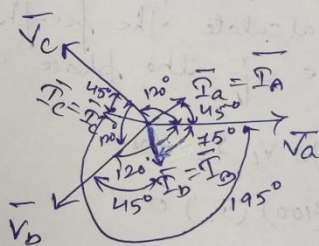
$$\text{Similarly } \bar{I}_b = 12.7 \angle -75^\circ \text{ A}$$

$$\bar{I}_c = 12.7 \angle -195^\circ \text{ A}$$

Since $\bar{I}_A = \bar{I}_a$, $\bar{I}_B = \bar{I}_b$, & $\bar{I}_C = \bar{I}_c$

$\therefore$ Magnitude (RMS) of phase & line current is 12.7 A

The phasor diagram can be drawn as.



Note:- Angles & lengths are not to scale in this figure.

Q.2 A balanced star connected load is fed by a star connected source. The source phase voltage is 200V and per phase load impedance is $100 \angle 60^\circ \Omega$. Calculate the magnitude of phase & line currents and the magnitude of the line-to-line voltage.

Solution-2

$$|V_p| = 200 \text{ V}$$

$$|V_{l-l}| = \sqrt{3} \times |V_p| = \sqrt{3} \times 200 = 346.4 \text{ V}$$

$$\text{Magnitude of line current} = \text{Magnitude of phase current}$$

$$= \frac{|V_p|}{|Z|} = \frac{200}{100} = 2 \text{ A}$$

P, Q, S, SPf

(4)

Q.3. For the problem given in Q.2, find (i) the active, and (ii) reactive power drawn by the load. Also calculate the power factor of the system.

Solution-3 $P = \sqrt{3} |V_L| |I_L| \cos \theta$
 $= \sqrt{3} (346.4) (2) \cos 60^\circ = 599.9 \approx 600 \text{ Watt}$

$$Q = \sqrt{3} |V_L| |I_L| \sin \theta$$
$$= (\sqrt{3}) (346.2) (2) \sin 60^\circ = 1039.2 \text{ VAR}$$

$$\text{Power factor} = \cos 60^\circ = 0.5 \text{ lagging}$$

Q.4. A three-phase delta connected load of 1 kW at 0.8 power factor lagging is fed from a three-phase source rated at 400 V. Calculate the per-phase impedance, and the magnitude of the phase and line currents.

Solution-4 $P = 1 \text{ kW} = \sqrt{3} V_L I_L \cos \theta$

$$\Rightarrow 1000 = \sqrt{3} (400) (I_L) 0.8$$

$$\Rightarrow I_L = \frac{1000}{\sqrt{3} \times 400 \times 0.8}$$

$$\Rightarrow I_L = 1.80 \text{ Ampere}$$

For delta connection $|I_L| = \sqrt{3} |I_p|$

$$\Rightarrow 1.80 = \sqrt{3} |I_p|$$

$$\Rightarrow |I_p| = 1.04 \text{ ampere}$$

Per-phase impedance

$$|V_p| = |I_p| |Z_p|$$

For delta $|V_p| = |V_L| = 400 \text{ V}$

$$\Rightarrow (400) = (1.04) |Z_p|$$

$$\Rightarrow |Z_p| = 384.61 \Omega$$

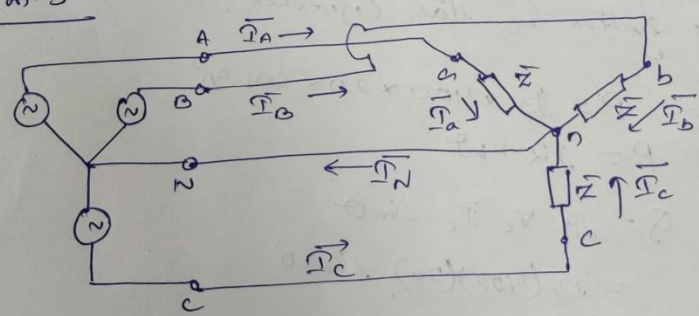
5

$pf = 0.8 \text{ lagging}$
 $\Rightarrow \theta_z = \cos^{-1} 0.8 = 36.86^\circ$

$\therefore \bar{Z}_p = |\bar{Z}_p| \angle \theta_z$
 $\bar{Z}_p = 384.61 \angle 36.86^\circ \Omega$

Q.5 A balanced three-phase star-connected load when fed from a symmetrical three-phase, 50 Hz, 400 V supply draws a current of 20 A in each phase such that the phase voltage leads the phase currents by 30° . Calculate the active & reactive power drawn by the load, per-phase load impedance and individual circuit components in each phase. Draw the phasor diagram of the system. If the source & load are connected by a three-phase four wire system, show that the current flowing in the neutral wire is zero.

Solution-5



Given $|V_L| = 400 \text{ V}$
 $\Rightarrow |V_p| = \frac{400}{\sqrt{3}} = 230.9 \approx 231 \text{ V}$

Take $\bar{V}_a$ (simply $\bar{V}_a$) as reference.

Then $\bar{V}_a = 231 \angle 0^\circ \text{ V}$
 $\bar{V}_b = 231 \angle -120^\circ \text{ V}$
 $\bar{V}_c = 231 \angle -240^\circ \text{ V}$

⑥

Given that voltage across each phase leads the phase current
 $\Rightarrow$ Phase current lags the phase voltage by 30°

$$\Rightarrow \bar{I}_a = |I_p| \angle -30^\circ$$

Given $|I_p| = 20 \text{ A}$

$$\Rightarrow \bar{I}_a = 20 \angle -30^\circ \text{ Ampere}$$

Similarly $\bar{I}_b = 20 \angle -150^\circ \text{ Ampere}$

& $\bar{I}_c = 20 \angle -270^\circ \text{ Ampere}$

Per-phase Impedance = $\frac{\bar{V}_a}{\bar{I}_a}$ (or any of the three phases since the system is balanced)

$$\bar{Z}_p = \frac{231 \angle 0^\circ}{20 \angle -30^\circ}$$

$$\bar{Z}_p = 11.55 \angle 30^\circ \Omega$$

3-phase active power = $\sqrt{3} V_L I_L \cos \theta$

Given: $|V_L| = 400 \text{ V}$

Since the system is star-connected, $|I_L| = |I_p| = 20 \text{ A}$

$$\Rightarrow P = \sqrt{3} \times 400 \times 20 \times \cos 30^\circ$$

$$P = 12 \text{ kW}$$

& $Q = \sqrt{3} V_L I_L \sin \theta$

$$= \sqrt{3} (400)(20) \sin 30^\circ$$

$$Q = 6.93 \text{ KVAR}$$

Individual circuit components:-

$$\bar{Z}_p = 11.55 \angle 30^\circ = 10 + j5.775 \Omega$$

$$\Rightarrow R = 10 \Omega$$

& $X = 5.775 \Omega$

$$\Rightarrow \omega L = 5.775$$

$$\Rightarrow L = \frac{5.775}{2\pi 50} = 18.3 \text{ mH}$$

$$\text{Neutral current, } \bar{I}_N = \bar{I}_a + \bar{I}_b + \bar{I}_c$$

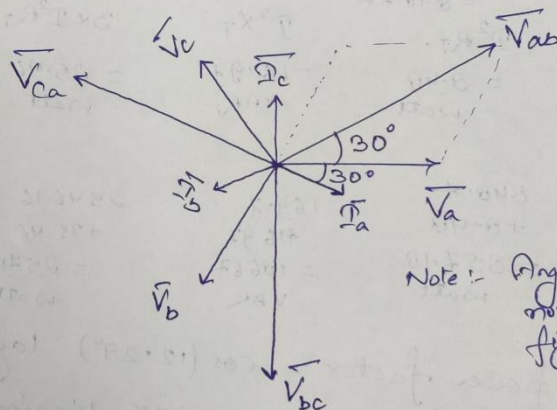
$$= 20 \angle -30^\circ$$

$$+ 20 \angle -150^\circ + 20 \angle -270^\circ$$

$$\bar{I}_N = 0 \text{ ampere.}$$

Thus, no current flows through the neutral circuit.

Phasor diagram (Take $\bar{V}_a$ as reference)



Note:- Angles & lengths are not to scale in this figure

Note: All angles are not written for the sake of clarity of the figure.

Three-phase circuit solution

Q.6 A balanced star connected load has a resistance of 50Ω & an inductor of 31.8 mH in series in each phase. The load is fed from a symmetrical three-phase source rated at 400 V through a symmetrical three-phase transmission line having a resistance of 0.5Ω in series with an inductor of 3.18 mH in each phase. The frequency of the source is 50 Hz . Calculate

the voltage at the load terminals, active & reactive power delivered to the load, active & reactive power lost in the transmission line, active & reactive power supplied by the source, current drawn by the system, and the resultant power factor of the system.

Solution-6 $f = 50 \text{ Hz}$

$$R_L = 50 \Omega, \quad L_L = 31.8 \text{ mH}$$

$$\Rightarrow X_L = \omega L = 2\pi \times 50 \times 31.8 \times 10^{-3} \approx 10 \Omega$$

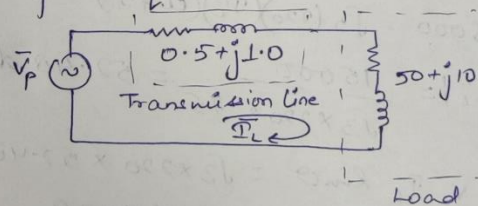
$$\therefore Z_L = R + jX_L = 50 + j10 \Omega$$

$$R_T = 0.5 \Omega, \quad L_T = 3.18 \text{ mH}$$

$$\Rightarrow X_T = 2\pi \times 50 \times 3.18 \times 10^{-3} \approx 1 \Omega$$

$$\therefore Z_T = R_T + jX_T = 0.5 + j1.0$$

Solve for any one phase
The per-phase equivalent can be drawn as:



$$|V_p| = \frac{|V_L|}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 231 \text{ V}$$

The total impedance in each phase, $\bar{Z}_p = (0.5 + j1) + (50 + j10)$

$$\bar{Z}_p = 50.5 + j11$$

$$\bar{Z}_p = 51.68 \angle 12.29^\circ \Omega$$

By taking the phase voltage as reference, $\bar{I}_L = \frac{V_p \angle 0^\circ}{\bar{Z}_p}$

$$\Rightarrow \bar{I}_L = \frac{231 \angle 0^\circ}{51.68 \angle 12.29^\circ} = 4.47 \angle -12.29^\circ \text{ Ampere}$$

(9)

Per-phase voltage across the load $= \bar{V}_{Load} = \bar{I}_L \times \bar{Z}_L$

$$\bar{V}_{Load} = (4.12 \angle -12.29^\circ) \times (50 + j10)$$

$$\bar{V}_{Load} = 210 \angle -0.98^\circ \text{ V}$$

$\therefore$ Magnitude of phase voltage at load terminals is 210 V
 $\Rightarrow$ Magnitude of line-to-line voltage at load terminals is $= \sqrt{3} \times 210 = 363.7 \text{ V}$

Power calculation :- Magnitude of phase current = 4.12 amperes

	Per phase		Total (3-phase)	
	Active	Reactive	Active	Reactive
Load	$I^2 R_L$ $= 4.12^2 \times 50$ $= 848.7 \text{ W}$	$I^2 X_L$ $= 169.7$ VAR	$3 \times I^2 R_L$ $= 2546.16$ Watt	$3 \times I^2 X_L$ $= 509.2 \text{ VAR}$
Line	$I^2 R_T$ $= 8.48$ Watt	$I^2 X_T$ $= 16.97$ VAR	$3 \times I^2 R_T$ $= 25.46$ Watt	$3 \times I^2 X_T$ $= 50.92$ VAR
Source	848.7 $+ 8.48$ $= 857.18$ Watt	169.7 $+ 16.97$ $= 186.67$ VAR	2546.16 $+ 25.46$ $= 2571.62$ Watt	509.2 $+ 50.92$ $= 560.12$ VAR

Resultant power factor $= \cos(12.29^\circ)$ lagging
 $= 0.977$ lagging.

(10)

Two wattmeter method

Q.7. A load having a power factor of 0.75 lagging draws 15 kW from a 3-phase, 50 Hz, 220 V source. If the power is measured by the two-wattmeter method, find the reading of the two wattmeters & the magnitude of line current.

Solution.7

$$P = 15000 = \sqrt{3} V_L I_L \cos \theta$$

$$15000 = \sqrt{3} (220) (I_L) (0.75)$$

$$|I_L| = \frac{15000}{\sqrt{3} \times 220 \times 0.75} = 52.48 \text{ A}$$

$$\therefore Q = \sqrt{3} V_L I_L \sin \theta = \sqrt{3} \times 220 \times 52.48 \times \sin(\cos^{-1} 0.75) \\ = 13227 \text{ kVAR}$$

In two wattmeter method

$$W_1 + W_2 = P \quad \& \quad W_1 - W_2 = Q/\sqrt{3}$$

$$\therefore W_1 + W_2 = 15000 \longrightarrow \textcircled{1}$$

$$\& \quad W_1 - W_2 = \frac{13227}{\sqrt{3}} = 7636.6 \longrightarrow \textcircled{2}$$

On solving equation ① & ②, we get,

$$W_1 = 11318 \text{ kW} \quad \& \quad W_2 = 3681.7 \text{ Watt}$$

(11)

alternatively, W_1 & W_2 can also be obtained without calculating ϕ , as follows:-

$$\phi = \cos^{-1} 0.75 = 41.4^\circ$$

$$W_1 = V_L I_L \cos(30^\circ - \phi) = 220 \times 52.48 \times \cos(30^\circ - 41.4^\circ)$$

$$\Rightarrow W_1 = 11.318 \text{ kW}$$

$$\begin{aligned} \text{f } W_2 &= V_L I_L \cos(30^\circ + \phi) = 220 \times 52.48 \times \cos(30^\circ + 41.4^\circ) \\ &= 3.682 \text{ kW} \end{aligned}$$

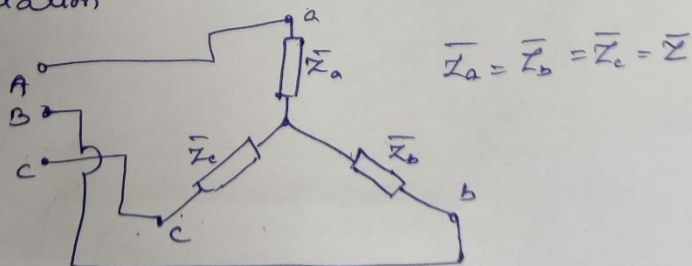
Star-Delta Conversion

Q.8. A balanced star connected load is fed from a three-phase, 50 Hz, 400 V supply. The per-phase impedance is $4.62 \angle 30^\circ \Omega$. Calculate the line current magnitude. Find an equivalent Δ representation of the load and show that the same line current is drawn by the equivalent delta connected load.

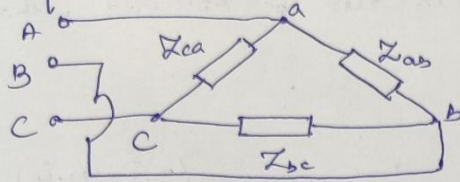
Solution-8 $|V_L| = 400 \text{ V} \Rightarrow |V_p| = \frac{400}{\sqrt{3}} = 231 \text{ V}$

$$|I_p| = |I_L| = \frac{|V_p|}{|Z|} = \frac{231}{4.62} = 50 \text{ A}$$

Star representation



(12)

Equivalent delta representation

$$\begin{aligned}\bar{Z}_{ab} &= \bar{Z}_{bc} = \bar{Z}_{ca} \\ &= \bar{Z}_\Delta = \frac{\bar{Z}_a \bar{Z}_b + \bar{Z}_b \bar{Z}_c + \bar{Z}_c \bar{Z}_a}{\bar{Z}_c} \\ \Rightarrow \bar{Z}_\Delta &= \frac{\bar{Z} \bar{Z} + \bar{Z} \bar{Z} + \bar{Z} \bar{Z}}{\bar{Z}}\end{aligned}$$

$$\Rightarrow \bar{Z}_\Delta = 3\bar{Z}$$

$$\Rightarrow \bar{Z}_\Delta = 3(4.62 \angle 30^\circ) \Omega$$

$$\text{Phase current magnitude} = \frac{|V_p|}{|Z_p|} = \frac{|V_{l-l}|}{|Z_\Delta|}$$

$$|I_p| = \frac{400}{3 \times 4.62} = 28.86 \text{ A}$$

In delta connected circuits, magnitude of line current = $\sqrt{3} \times |I_p|$ [Refer Q.1]

$$= \sqrt{3} \times 28.86$$

$$|I_L| = 49.4 \approx 50 \text{ Amp [Same as the star representation]}$$

